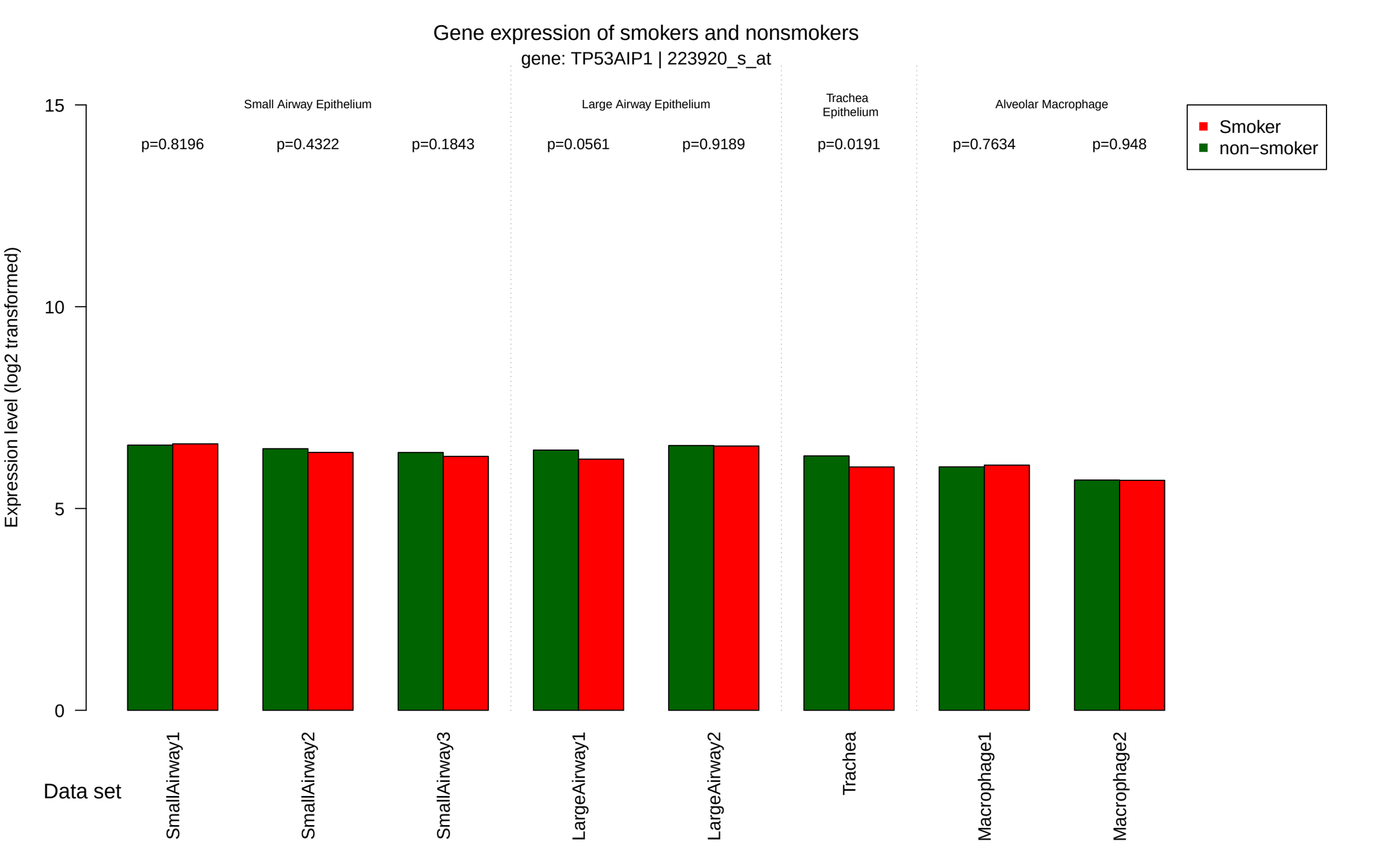
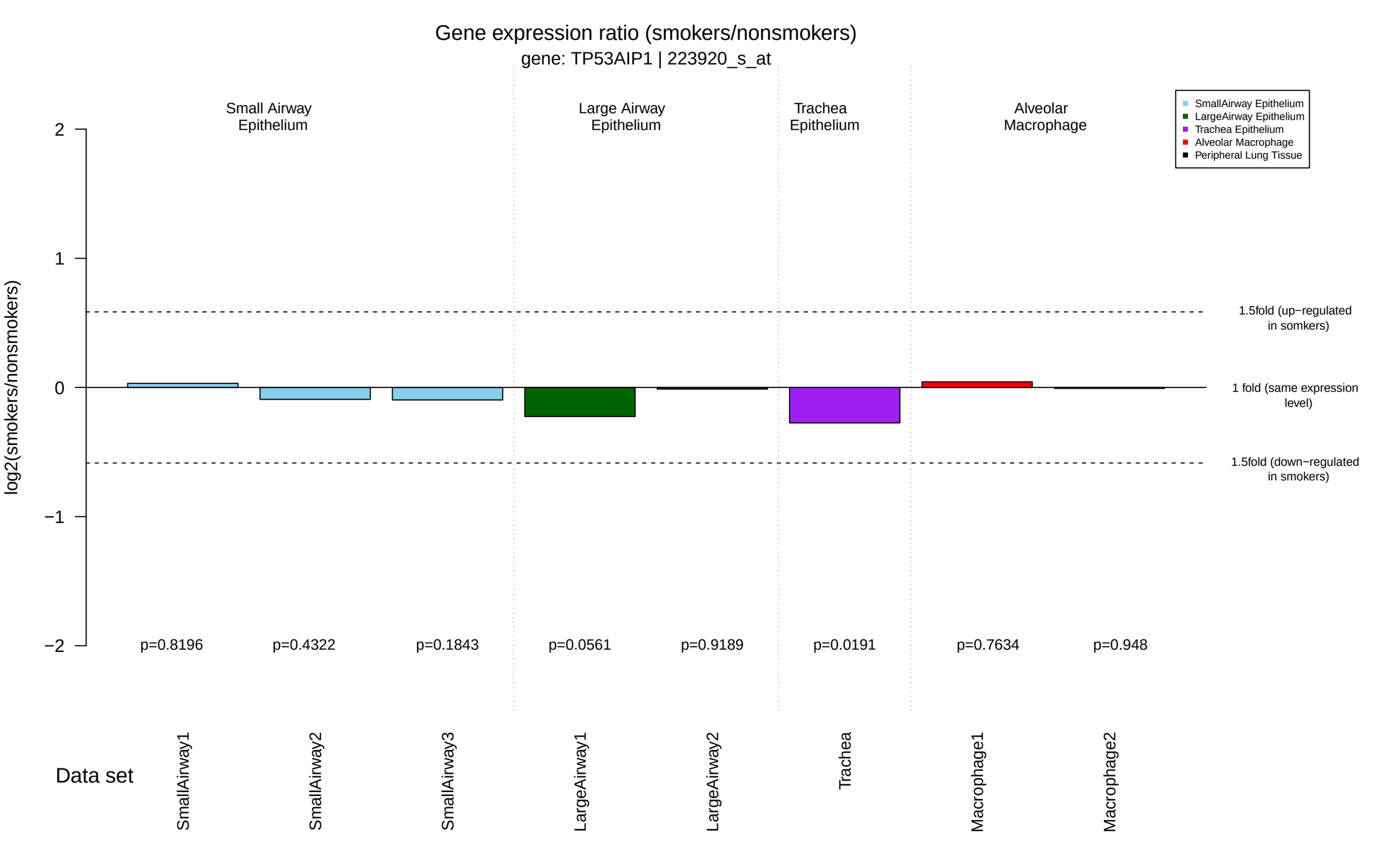
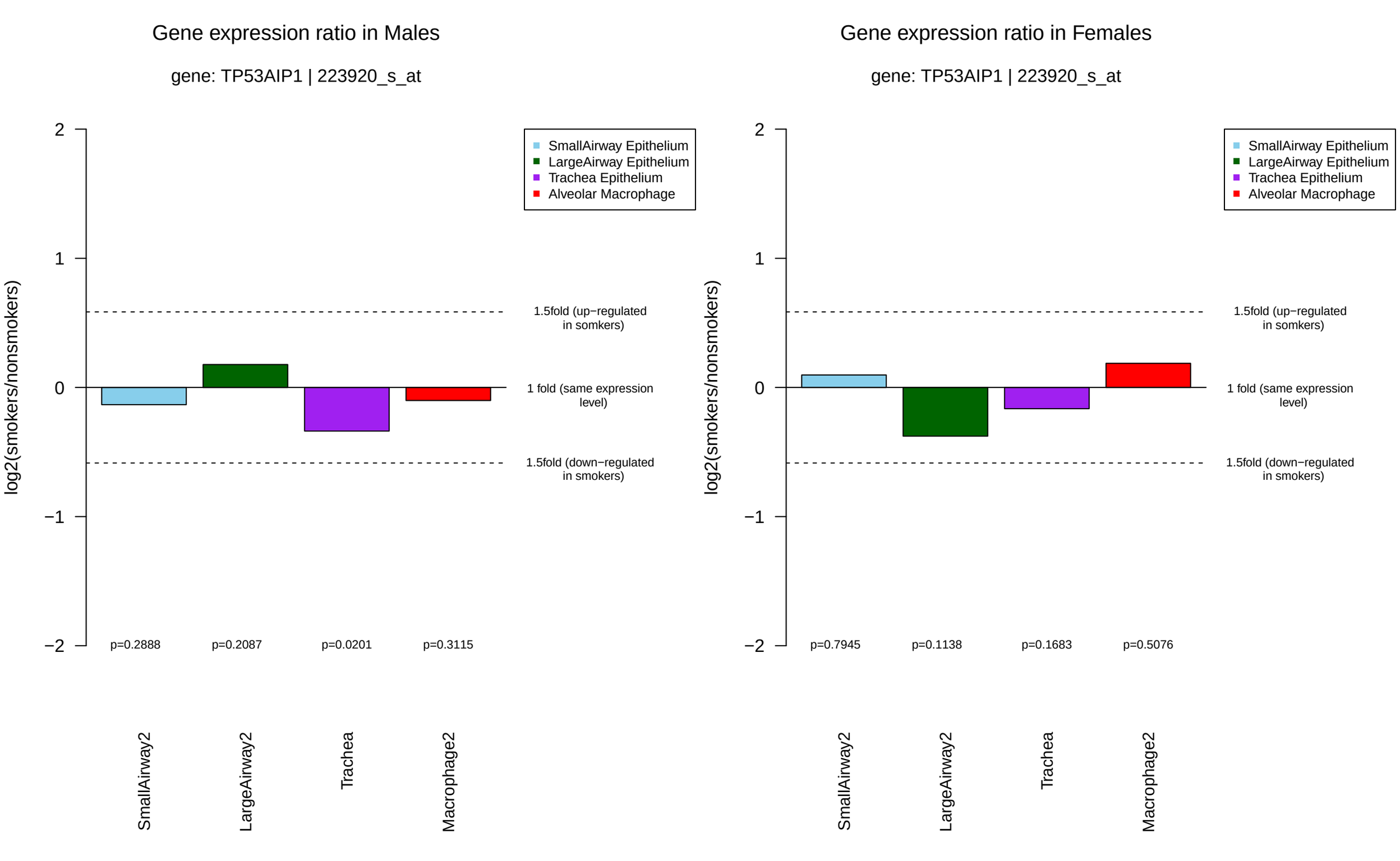
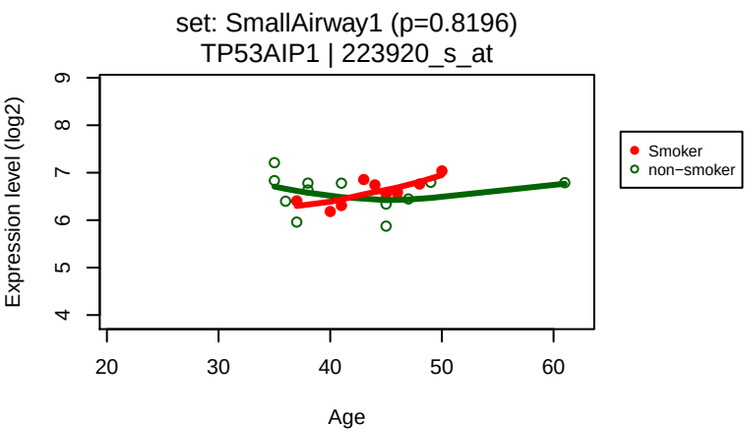
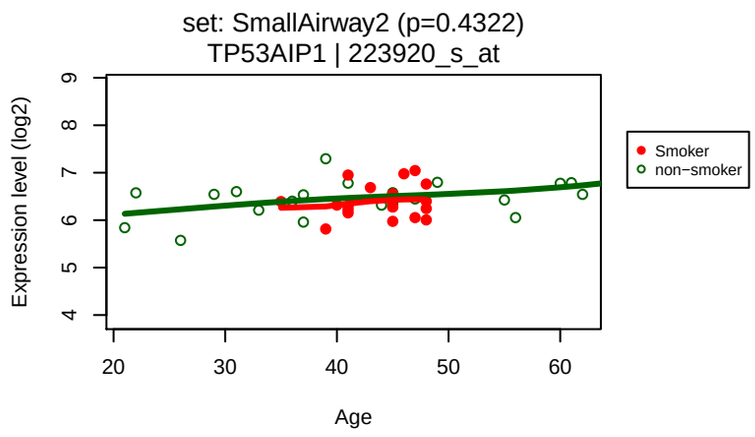
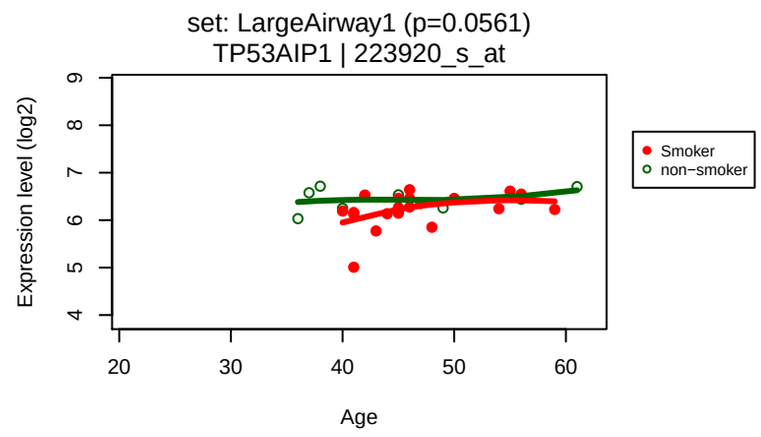

Gene expression of smokers and nonsmokers
gene: TP53AIP1 | 223920_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8196
p=0.4322
p=0.1843
p=0.0561
p=0.9189
p=0.0191
p=0.7634
p=0.948
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

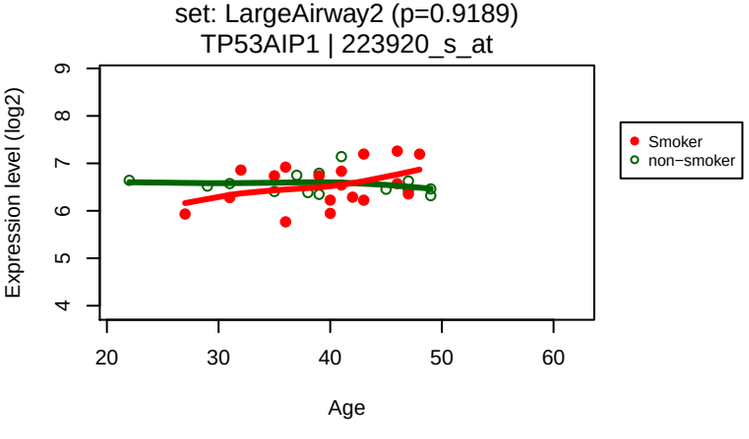

Gene expression ratio (smokers/nonsmokers)
gene: TP53AIP1 | 223920_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8196
p=0.4322
p=0.1843
p=0.0561
p=0.9189
p=0.0191
p=0.7634
p=0.948
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

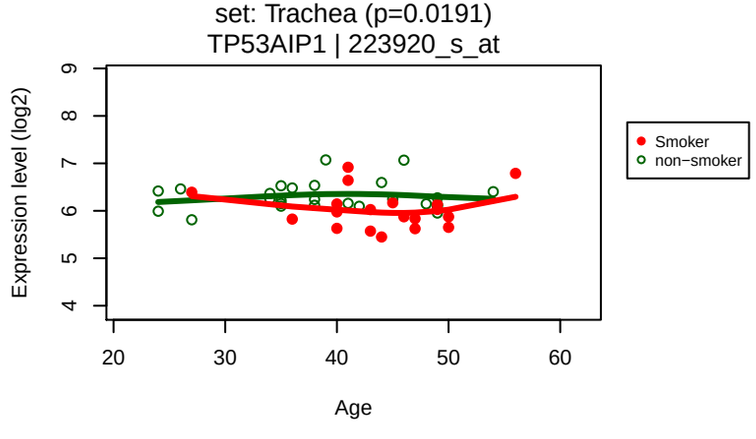

Gene expression ratio in Males
gene: TP53AIP1 | 223920_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2888
p=0.2087
p=0.0201
p=0.3115
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TP53AIP1 | 223920_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7945
p=0.1138
p=0.1683
p=0.5076
SmallAirway2
LargeAirway2
Trachea
Macrophage2

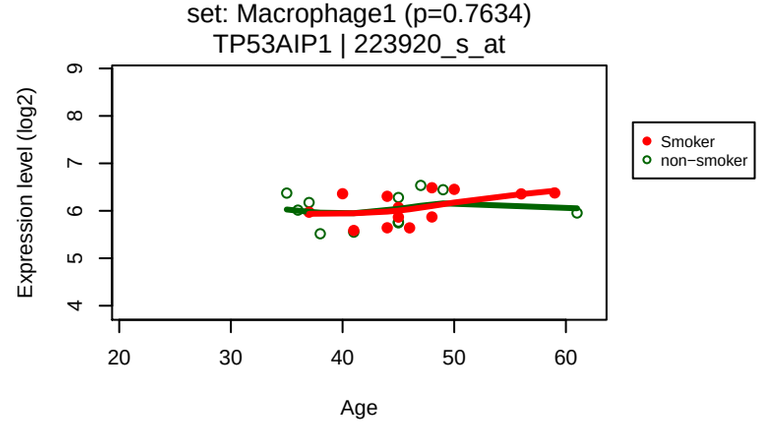

set: SmallAirway1 (p=0.8196)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker

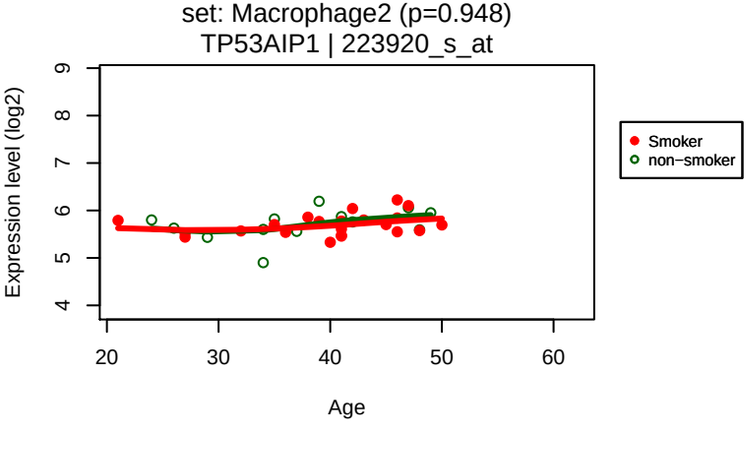

set: SmallAirway2 (p=0.4322)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker

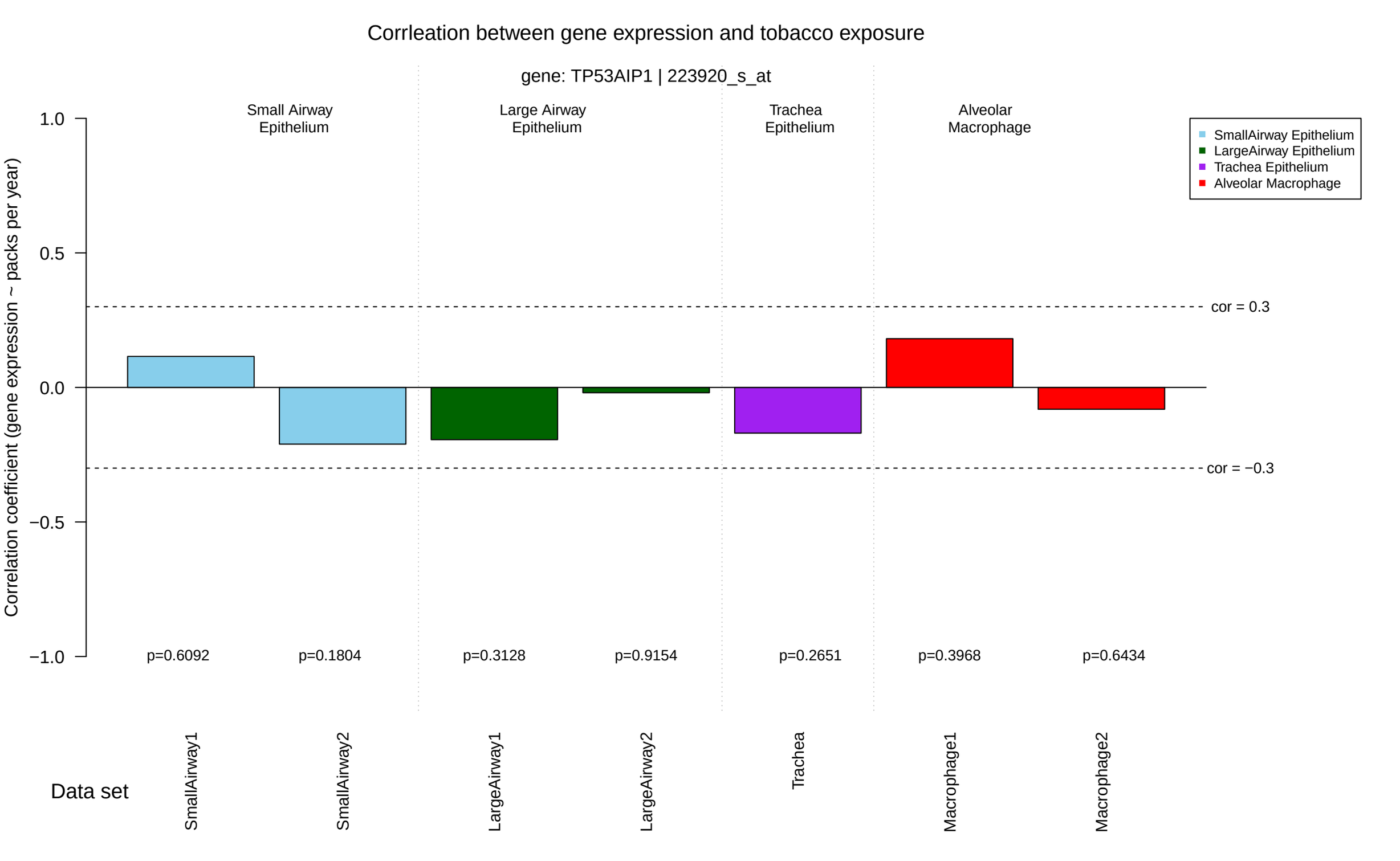

set: LargeAirway1 (p=0.0561)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9189)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0191)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7634)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.948)
TP53AIP1 | 223920_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TP53AIP1 | 223920_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6092
p=0.1804
p=0.3128
p=0.9154
p=0.2651
p=0.3968
p=0.6434
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2